LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Overall consideration of aircraft loading action.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Overall consideration of aircraft loading action.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of structural element diagrams from IB; Instructor supervision during Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary functions of the aircraft structure?
(b) List three major structural elements of a typical aircraft.
(c) How does the fuselage contribute to overall aircraft integrity?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of aircraft loading action • Types of structural loads • Material properties in structure • Load distribution principles • Safety factors in design	L & Ds	PPT, IB	15 min
(b)	• Analysis of symmetric flight loads • Impact of maneuvers on structure • Stress concentration points • Fatigue life considerations • Maintenance inspection criteria	L & Ds	PPT, PDF	20 min
(c)	• Integration of structural elements • Role of fasteners and joints • Corrosion prevention methods • Repair techniques for damaged areas • Documentation of structural repairs	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how different structural elements work together to handle loading actions.
(b) Describe the importance of safety factors in aircraft structural design.

12. Home Assignment and Follow Up	Prepare a brief report on how the principles discussed in this lesson apply to the next lesson on symmetric flight loads.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Symmetric flight loads, symmetric maneuvers	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
John Doe, Flight Lieutenant, P No. 12345	Jane Smith, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min